TASK 1: RESEARCH REPORT ON ROBOTICS APPLICATIONS

Robotics in Healthcare: Surgical Robots, Rehabilitation and Intelligent Patient Support

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Abstract

Healthcare robotics combines mechanical systems, sensors, control and software to support clinicians and patients. Major applications include surgical robots, rehabilitation robots, hospital logistics and assistive systems. This report discusses their working principles, benefits, limitations and future scope.

1. Introduction

Healthcare requires precision, repeatability and safety. Robots can assist with minimally invasive procedures, repeated rehabilitation movements and routine logistics. They support human professionals rather than automatically replacing clinical decision-making.

2. Surgical Robotics

A surgical robotic system typically contains a surgeon interface, robotic manipulators, instruments, cameras and a control computer. The surgeon provides commands while the system translates them into controlled instrument movements. Potential benefits include precise and stable motion, but suitability depends on the procedure, equipment, training and clinical evidence.

3. Rehabilitation Robotics

Rehabilitation robots support repeated therapeutic movements. Exoskeletons and upper- or lower-limb devices can use encoders, force sensors and motion sensors to measure movement. Controllers can adjust assistance according to a therapy plan and record progress.

4. Assistive and Service Robots

Assistive robots can help people with mobility or daily tasks. Hospital service robots can transport medicines, samples and supplies. These robots need dependable perception and safe human-robot interaction because hospital environments change continuously.

5. Sensors and Control

Cameras provide visual information; encoders measure joint position; IMUs estimate movement; proximity sensors detect nearby objects; and force/torque sensors can detect contact. Closed-loop control repeatedly senses the robot state, compares it with the desired state, acts and measures the result.

6. Role of AI

AI can improve perception, object recognition, prediction, planning and adaptation. Machine learning and deep learning can process complex sensor data. New research also explores language and foundation models for planning and interaction. In healthcare, AI should remain subject to validation, monitoring and human oversight.

7. Benefits

Benefits include improved precision and repeatability, support for tiring or hazardous tasks, personalized rehabilitation, structured performance data and automation of routine hospital movement.

8. Challenges

Challenges include cost, maintenance, cybersecurity, privacy, reliability, training and integration with hospital workflows. AI introduces additional risks such as prediction errors, bias, limited explainability and unexpected behavior.

9. Future Scope

Future systems may combine richer sensors, safer learning-based control, better human-robot interaction, autonomous hospital navigation and multi-robot coordination. Safety constraints and human supervision will remain essential.

10. Conclusion

Healthcare robotics is a combination of mechanics, sensing, control and software, increasingly enhanced by AI. Surgical, rehabilitation and service robots demonstrate how automation can support healthcare while keeping humans central to safe decisions.

References

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